ASTANA IT UNIVERSITY

«Approved»
by the Dean
Syzdykova Z.A.
«___» of 06 2024

Syllabus

Academic Year 2024 -2025

1. General information	
Course title	Cloud Computing
Degree cycle (level)/ major	6B06103 – Big Data Analysis 6B06106 – Mathematical and Computational Science
Year, term	3, 7-8
Number of credits	5
Language of delivery:	English
Prerequisites	"Cloud computing" is based on knowledge obtained by the students on Information and Communication Technologies, Web technologies, Database Management Systems, Computer Organisation and Architecture, and Operating Systems and Computer Networks courses.
Postrequisites	Knowledge obtained during the course will be used for "Network Application Testing" and "Software Quality Assurance and Testing" courses.
Lecturer(s)	Aldiyar Salkenov, Master of Engineering, Senior-lecturer, aldiyar.salkenov@astanait.edu.kz, C1.3.357. Office hours: Saturday, 13:00
2. Goals, objectives and learning outcomes of the course	
1. Course description	This course introduces you to the core concepts of cloud computing. You will gain the foundational knowledge required for understanding cloud computing from practitioner perspective. You will learn about the definition and essential characteristics of cloud computing, its history, emerging trends, and the business case for cloud computing. You also learn about the various cloud service models (IaaS, PaaS, SaaS) and deployment models (Public Cloud, Private Cloud, Hybrid Cloud) and the key components of a cloud architecture (Virtualization, VMs, Storage, Networking, Containers). The course will also familiarize you with emerging trends associated with cloud including, Microservices, Serverless, Cloud Native, DevOps, and Application Modernization. You learn about Cloud Security basics and will be introduced to some case studies and career opportunities related to cloud computing.
2. Course goal(s)	This course is designed to give students a detailed overview of core cloud services, security, architecture, pricing, and support. Study of the technical and economic feasibility of transferring existing applications to the cloud; familiarity with the cloud computing infrastructure; scaling, deployment, backup, and solving security issues in the context of cloud infrastructure.

3. Course objectives:	You will learn: 1. Essential characteristics and benefits of Cloud Computing, e.g. pay-per-use, etc. 2. Top Cloud Service Providers 3. Common cloud service models (IaaS, PaaS, SaaS) and deployment models (Public, Private, Hybrid). 4. Components of cloud infrastructure (Regions, Availability Zones, Data Centers, Virtualization, VMs, Bare Metal, Networking), and types of cloud storage (Direct Attached / Ephemeral, Persistent - File Storage, Block Storage, Object Storage, etc.) 5. Emergent trends in cloud computing - including Hybrid MultiCloud, Containers, Microservices, Serverless, Cloud Native, DevOps, and Application Modernization. 6. Concepts in cloud security, encryption and monitoring. 7. Career options, cloud related professions, and their required skills.
4. Skills & competences	To successfully pass this course you should have prior knowledge and skills on: - web technologies (web servers and services); - computer networks (IP, DNS, CIDR, etc.); - operating systems and command line interface (Linux, bash, etc.);
5. Course learning outcomes:	After completing this course, you will be able to: – Define cloud computing and explain its essential characteristics, evolution, and business case for cloud adoption; – Define the cloud service models (IaaS, PaaS, SaaS) and deployment models (Public, Private, Hybrid); – Describe the concepts and components of cloud infrastructure such as virtualization, virtual machines, virtual network, bare metal servers, and containers; – Explain different cloud storage models including Direct Attached, File, Block, Object Storage, and CDNs; – Explain emergent trends related to cloud computing including Microservices, Serverless computing, Cloud Native, DevOps, and Application Modernization; – Describe the concepts in cloud security and how organizations in different industries are benefiting from cloud; – List the job roles and career opportunities available in cloud computing; – Prepare for the AWS Certified Cloud Practitioner exam;
6. Methods of assessment	– Graded quizzes on lecture materials. – Lab assignments in the AWS Academy Learner Lab environment. – Self-study on AWS Academy Cloud Foundations course.
7. Reading list	1. "Amazon Web Services in Action" by M. Wittig, A. Wittig (2016) 2. "Explain the Cloud Like I'm 10" by Todd Hoff (2017) 3. "Ahead in the Cloud: Best Practices for Navigating the Future of Enterprise IT" by S. Orban et al. 1st edition (2018), 334 p. 4. The NIST Definition of Cloud Computing, NIST Special Publication 800-145. 5. International Standard ISO/IEC 22123-1-2021 Information technology — Cloud computing — Part 1: Vocabulary

8. Resources	1. https://www.awsacademy.com/ 2. https://docs.microsoft.com/en-us/learn/ 3. https://cloud.google.com/
4. Course policy	**Course and university policies include:** **Attendance:** Students are expected to attend all scheduled class sessions with all required reading and supplementary materials. Readings are to be completed prior to class. The student won't obtain additional points for course attendance, but attendance is important to pass the course. In case the student is not able to attend the classes for some reasons, he/she must inform the dean's office in advance and the student itself is responsible for learning all materials, which were given during unattended lessons. In case if **the student did not attend more than 30% of the classes** without any reasonable excuses, the teacher has a right to mark him as "not graded", and the student wouldn't be admitted to the exam. In other words, students must participate in at least **70%** of all class time, otherwise he/she fails the course. **Preparation for Class: Class participation is an important part of this course's learning process.** Although not explicitly graded, students will be evaluated on the QUALITY of their contributions and insights. Quality comments possess one or more of the following properties: - Offers a different and unique, but relevant, perspective; - Contributes to moving the discussion and analysis forward; - Builds on other comments. **Class work**: The duration of each lecture and practical lesson is 50 minutes for offline class, and 40 minutes for online class. Students are expected to complete all readings and assignments ahead of time, attend class regularly and participate in class discussions. In case of systemic student misconduct, the student would be dispensed from the classes. **Being late to class:** When students come to class late, it can disrupt the flow of a lecture or discussion, distract other students, impede learning, and generally erode class morale. Moreover, if left unchecked, lateness can become chronic and spread throughout the class. By the policy of this course, students who come late to class for more than 5 minutes are not allowed to get in to class and consequently, they will be marked as "absent" for the specific hour. **Attestation I and II:** Students with a score of less than 25% for Attestation period I or Attestation period II (RK1/RK2) automatically failed and should take the course again. **Home work / Assignments**: The assignments are designed to acquaint students with the theoretical knowledge and practical skills required for the course. The textbook readings will be supplemented with materials collected from recent professional articles and journals. In case of using

	someone's work (papers, articles, any publications), all works must be properly cited. Failure to cite work will result as cheating from the students and may be subject of additional disciplinary measures. **Late assignments:** Most assignments will be discussed in class on the due date, therefore late assignments will not receive credit. All work is expected to be submitted on time. Failure to pass assignments on time will result in 0% for the assignment. In other words, no late submissions are allowed. All gradings are based using a percentage grading scale. *In some extraordinary event, students should notify the teacher and request an extension of the deadline. If approved, a new date will be given to the student depending upon the circumstances.* **Final exam:** The final exam will be held according to the schedule in the Academic Calendar. It will be in a computer testing form. **Laptops and mobile devices** can only be used for classroom purposes when directed by the teacher. Misuse of laptops or handheld devices will be considered a breach of discipline and appropriate action will be initiated by the teacher. **Online lessons** can be used in case there won't be a chance to make offline traditional lessons. It must not discourage the interest and enthusiasm of students. The main software to run the online lessons is Microsoft Teams for video calls and live webinars, and Moodle (moodle.astanait.edu.kz) as a Learning Management System. Also, some alternatives such as Zoom or Telegram messenger may be involved as an additional workaround. **Cheating and plagiarism** are defined in the Academic conduct policies of the university and include: 1. Submitting work that is not your own papers, assignments, or exams; 2. Copying ideas, words, or graphics from a published or unpublished source without appropriate citation; 3. Submitting or using falsified data; 4. Submitting the same work for credit in two courses without prior consent of both instructors. Any student who is found cheating or plagiarizing on any work for this course will receive 0 (zero) for that work and further actions will also be taken regarding academic conduct policies of the university. **Academic Conduct Policies of the university:** The full texts of all the academic conduct code will be posted to the students using Learning Management System (moodle.astanait.edu.kz). **Contacting the Instructor (Teacher)**: The easiest and most reliable way to get in touch with the teacher is by email. Students must feel free to send an email if you have a question related to the course. The teachers will respond as soon as they can but not always instantaneously. Besides that, students are also welcome to arrange a

	one-to-one meeting with the teacher by their office during office hours to discuss the class using both offline and online ways.

3. Course Content

#	Abbreviation	Meaning
1	TSIS	Teacher-supervised independent work
2	SIS	Students' independent work
3	IP	Individual project
4	PA	Practical assignment
5	LW	Laboratory work
6	MCQ	Multiple choice quiz

3.1 Lecture, practical/seminar/laboratory session plans

Week No	Course Topic	Lectures (H/W)	Practice sessions (H/W)	Lab. sessions (H/W)	TSIS (H/W)	SIS (H/W)
1	- Cloud Concepts Overview - Cloud Economics and Billing	3	2	0	0	9
2	- AWS Global Infrastructure Overview - AWS Cloud Security	3	2	0	1	9
3	- Networking and Content Delivery	3	2	0	1	9
4	- Compute	3	2	0	2	9
5	- Storage - Databases	3	2	0	1	9
6	- Cloud Architecture - Auto Scaling and Monitoring	3	2	0	1	9
7	- AWS Introduction to Machine Learning (ML)	3	2	0	1	9
8	- Implementing a ML pipeline with Amazon SageMaker (Part 1)	3	2	0	1	9
9	- Implementing a ML pipeline with Amazon SageMaker (Part 2)	3	2	0	1	9
10	- Introduction to Forecasting - Review	3	2	0	1	9
	Total hours: 150	**30**	**20**	**0**	**10**	**90**

3.2 List of assignments for Student Independent Study

№	Assignments (topics) for Independent study	Hours	Recommended literature and other sources (links)	Form of submission
1	AWS Cloud Foundations. Module 1: Cloud Concepts Overview Module 2: Cloud Economics and Billing	9	AWS Academy Cloud Foundations course. Teachers will enroll you to the course by sending invitation email. You	Teacher can see your progress and grades in AWS

2	Module 3: AWS Global Infrastructure Overview Module 4: Cloud Security	9	must follow the link in the email and create an account in AWS Academy LMS. Start learning using navigation menu in the left side of LMS, watching videos, reading guides, doing labs, and passing knowledge check quizzes.	Academy LMS. You must complete Pre-Course and End of Course Feedback Surveys.
3	Module 5: Networking and Content Delivery	9		
4	Module 6: Compute	9		
5	Module 7: Storage Module 8: Databases	9		
6	Module 9: Cloud Architecture Module 10: Automatic Scaling and Monitoring	9		
7	AWS Machine Learning Foundations Module 2: Introducing Machine Learning	9		
8	Module 3: Implementing a ML pipeline with Amazon SageMaker	9		
9	Module 3: Implementing a ML pipeline with Amazon SageMaker	9		
10	Module 4: Introducing Forecasting	9		

4. Student performance evaluation system for the course

Period	**Assignments**	**Number of points**	**Total**
1st attestation	Assignment 1 (AWS Academy labs) Assignment 2 (AWS Academy labs) Quiz 1 **Midterm quiz**	30 30 10 **30**	100
2nd attestation	Assignment 3 (AWS Academy labs) Assignment 4 (AWS Academy labs) Quiz 2 **Endterm quiz**	30 30 10 **30**	100
Final exam	Multiple choice quiz		100
Total	**0,3 * 1st Att + 0,3 * 2nd Att + 0,4*Final**		**100**

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

Letter Grade	**Numerical equivalent**	**Percentage**	**Grade according to the traditional system**
A	4,0	95-100	Excellent
A-	3,67	90-94	
B+	3,33	85-89	Good
B	3,0	80-84	
B-	2,67	75-79	
C+	2,33	70-74	
C	2,0	65-69	Satisfactory
C-	1,67	60-64	
D+	1,33	55-59	

D	1,0	50-54	
FX	0	25-49	Fail
F	0	0-24	

Based on the specific grade for each assignment, and the final grade, following criteria must be satisfied:

Grade	Criteria to be satisfied
90-100	- Understand the core concepts of the cloud computing paradigm: how and why this paradigm shift came about, the characteristics, advantages and challenges brought about by the various models and services in cloud computing - Analyze various cloud programming models and apply them to solve problems on the cloud - Ability to work with multi-tier architectures
80-89	- Excellent range and depth of attainment of cloud computing - Mastery of a wide range of AWS services - Evidence of study and originality of what has been practiced - Able to define the global infrastructure components of AWS
70-79	- Attained all the intended learning outcomes for AWS Cloud architectural principles - Able to use well a range of methods and techniques to come to conclusions - Able to employ critical analysis and judgement
60-69	- Some limitations in attainment of learning objectives, but has managed to grasp most of AWS Services - Able to use most of the AWS Services - Some grasp of the issues and concepts underlying global infrastructure components of AWS, but weak and incomplete
50-59	- Attainment of only a minority of the learning outcomes - Able to demonstrate a clear but limited use of some of the AWS Cloud - Weak and incomplete grasp of AWS components - Deficient understanding of the issues and concepts underlying the techniques and materials
25-49	- Attainment of nearly all the intended learning outcomes deficient - Lack of ability to use at all or the right AWS components - Inadequately and incoherently presented - Wholly deficient grasp of AWS services - Lack of understanding of the issues and concepts underlying AWS Cloud
0-24	No significant assessable material, absent or assessment missing a must pass component

5. Methodological Guidelines

Assessment is administered continuously throughout the course. The students are rated against their performance in continuous rating administered throughout the semester (60%) and summative rating done during the examination session (40%), total 100%. Continuous rating is students' on-going performance in class and independent work. Class work is assessed for attendance, laboratory works' defense and in- class assessments.

- **TSIS (Teacher Supervised Student Independent Study)** -comprises presentation to be done by students independently and checked by instructor.
- **Mid-term and End-term** is a review of the topics covered and assessment of each student's knowledge. The form of the midterm and end term exams is MCQ.

- **Final assessment**: The final exam will be held according to schedule in the Academic Calendar. The form of the final exam is computer testing.

6. Lecturer (lecturers) approvals Full name Job title Date Sign

Full name	Job title	Date	Sign
Aldiyar Salkenov	Senior Lecturer	02.09.2024	

Director of Department of Computer Engineering **Praveen Kumar**